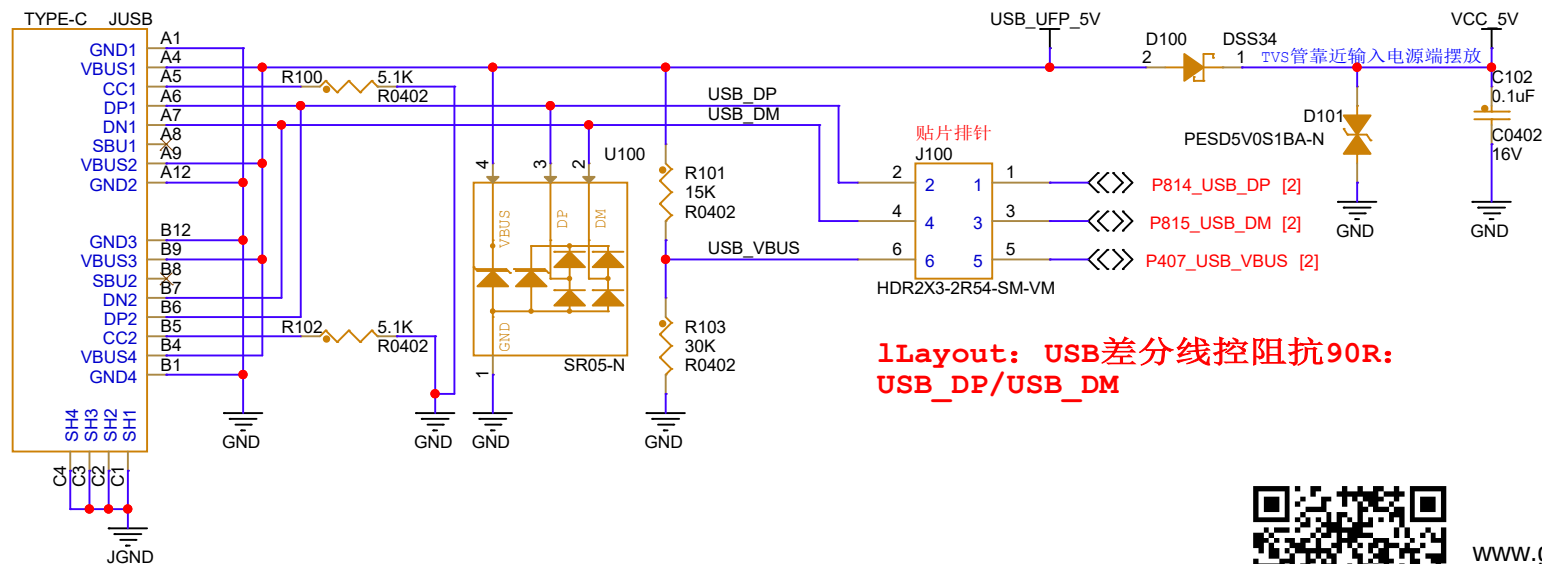
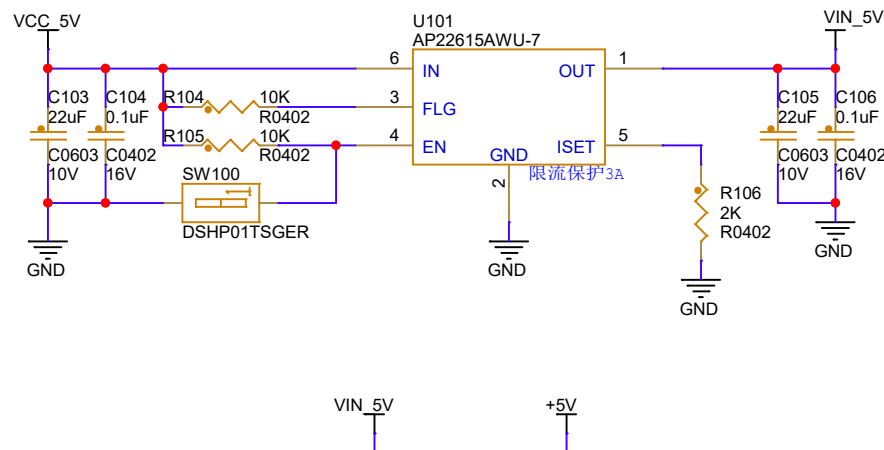




1Layout: USB差分线控阻抗90R:
USB DP/USB DM



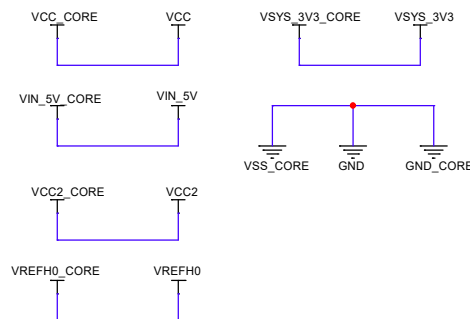
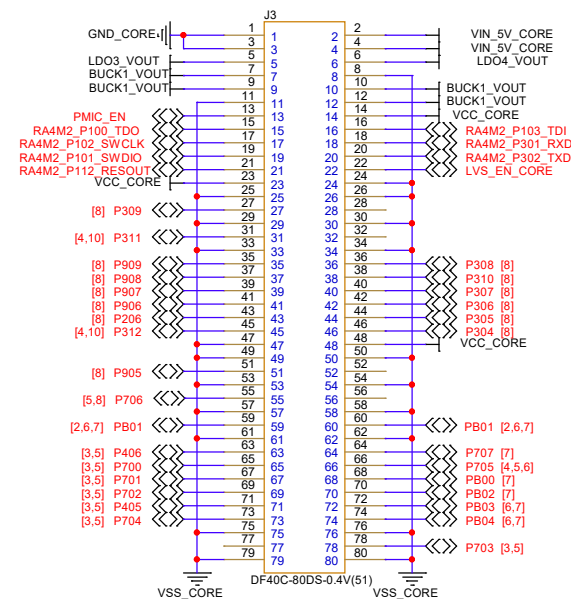
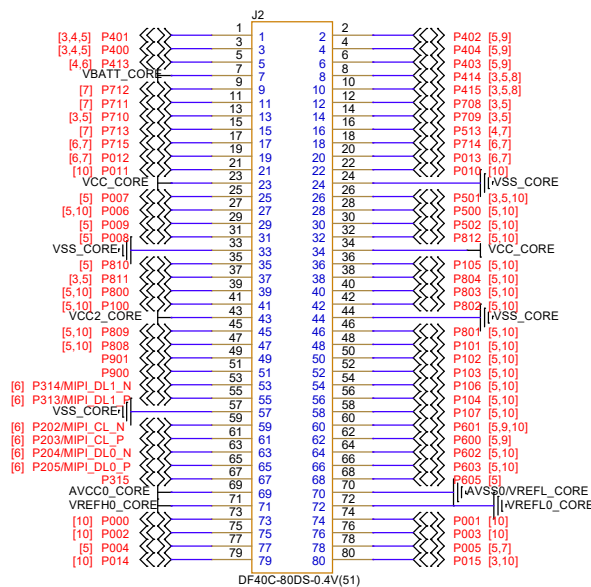
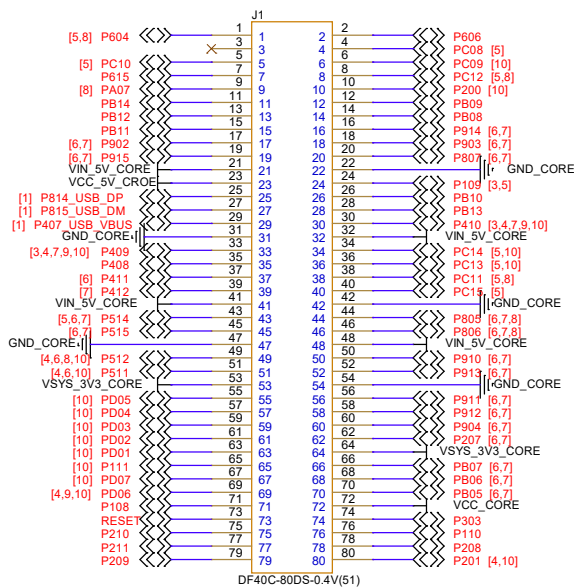
[www.gitee.com/
ramcu/
cpk_examples](https://www.gitee.com/ramcu/cpk_examples)



[www.github.com/
renesas/
cpk_examples](https://www.github.com/renesas/cpk_examples)

BTB座子

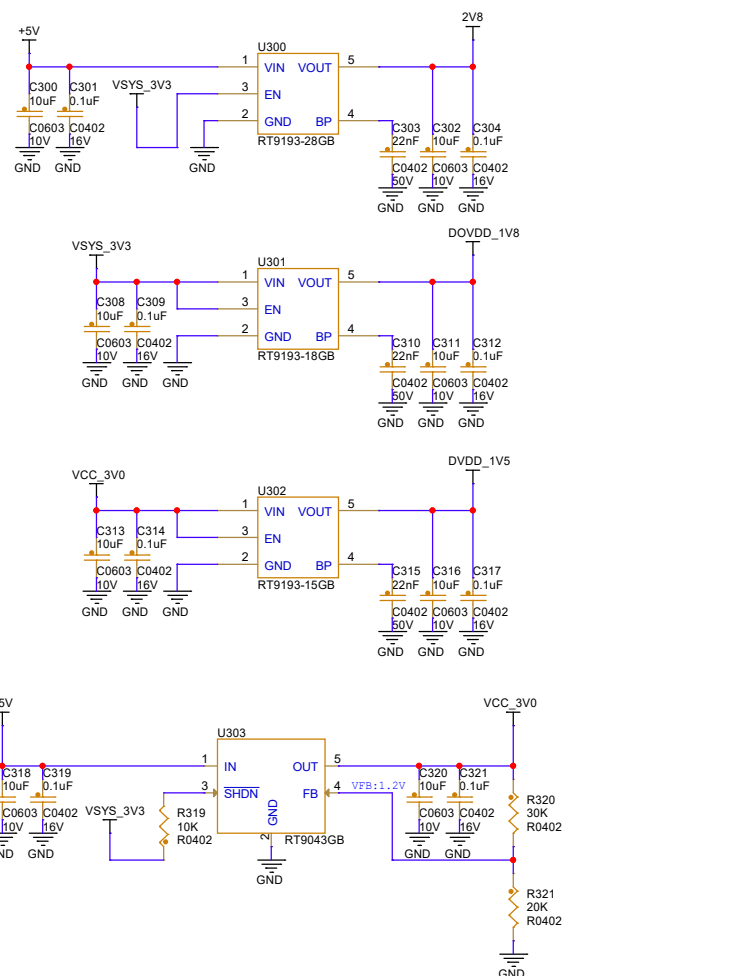
→ Page1: USB_FS
Page2: BTB
Page3: OV/MIPI
Page4: CAN/陀螺仪
Page5: PMOD
Page6: RGB/MIPI LCD
Page7: RGB转HDMI/HDMI接口
Page8: 千兆网口
Page9: 音频PDM
Page10: Arduino



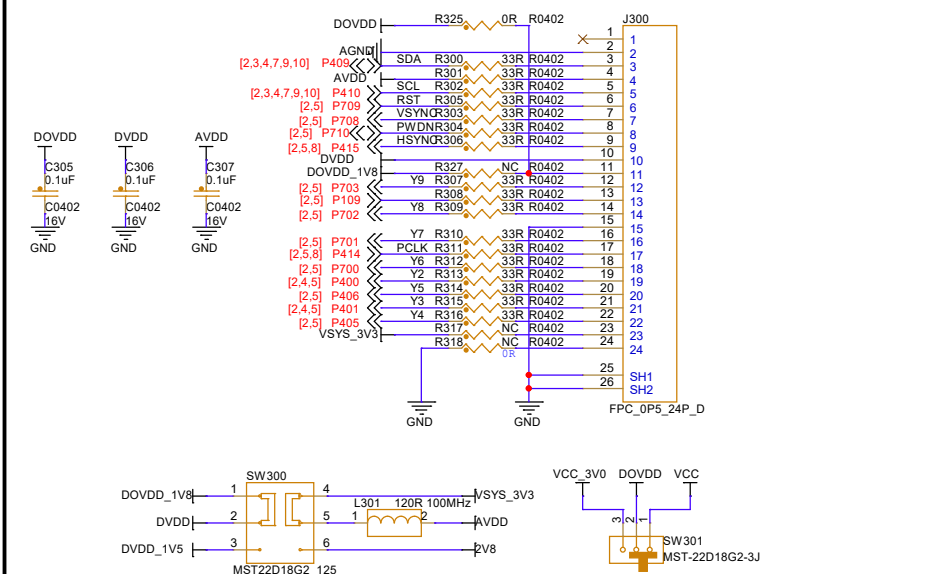
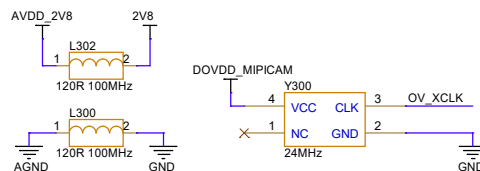
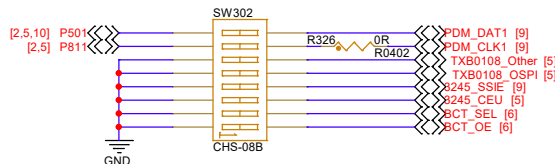
定位孔



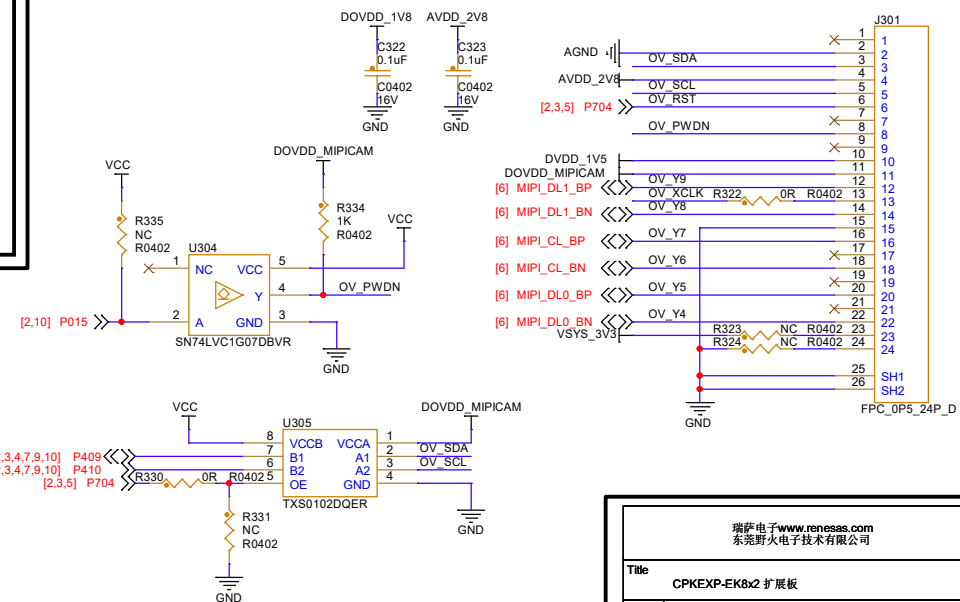
瑞萨电子www.renesas.com 东莞野火电子技术有限公司			
Title CPKEXP-EK8x2 扩展板			
Size A3	Document Number BTB		Rev V1
Date:	Saturday, September 12, 2026	Sheet 2	of 12



拨码开关



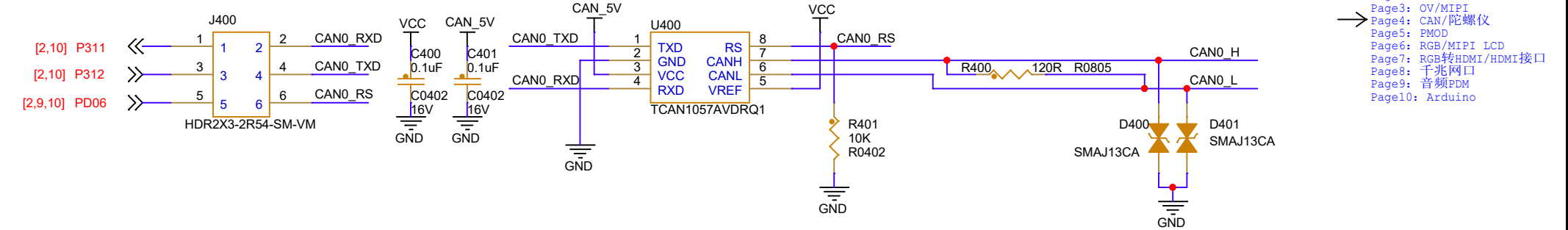
MIPI DSI



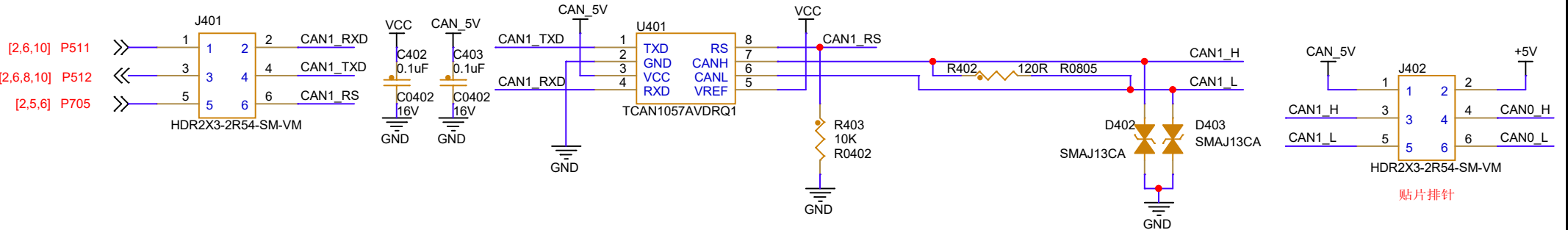
→

- Page1: USB_FS
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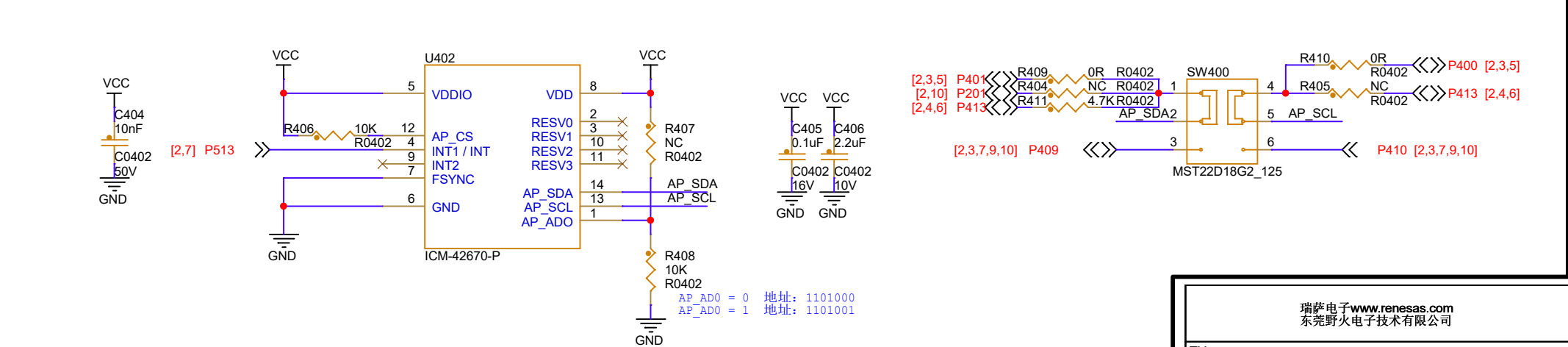
CAN0



CAN1

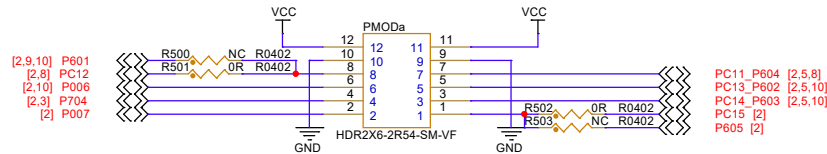


陀螺仪

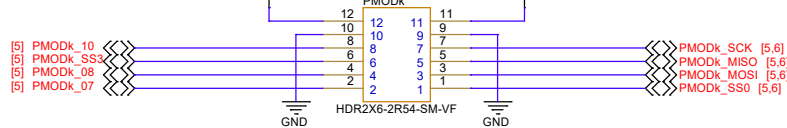


瑞萨电子www.renesas.com 东莞野火电子技术有限公司		
Title CPKEXP-EK8x2 扩展板		
Size A4	Document Number PDM/CAN	Rev V1.0
Date: Saturday, September 12, 2026	Sheet 4	of 12

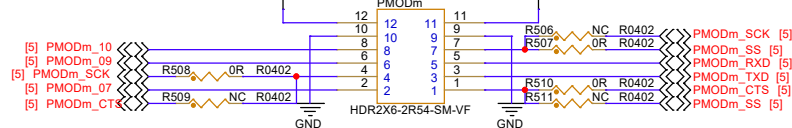
PMODa



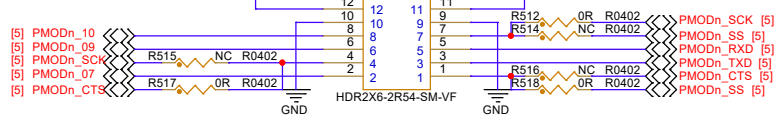
PMODk



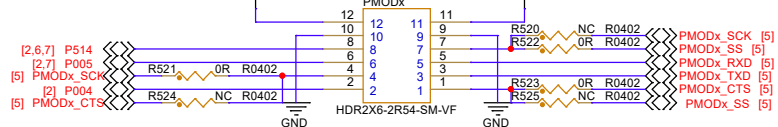
PMODm



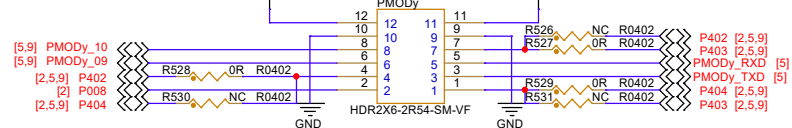
PMODn



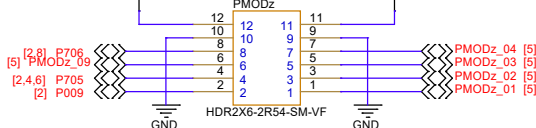
PMODx



PMODY

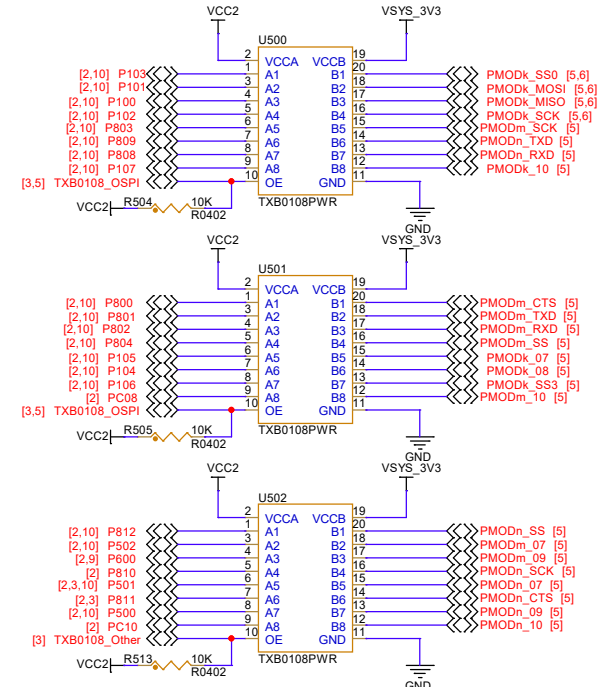


PMODz



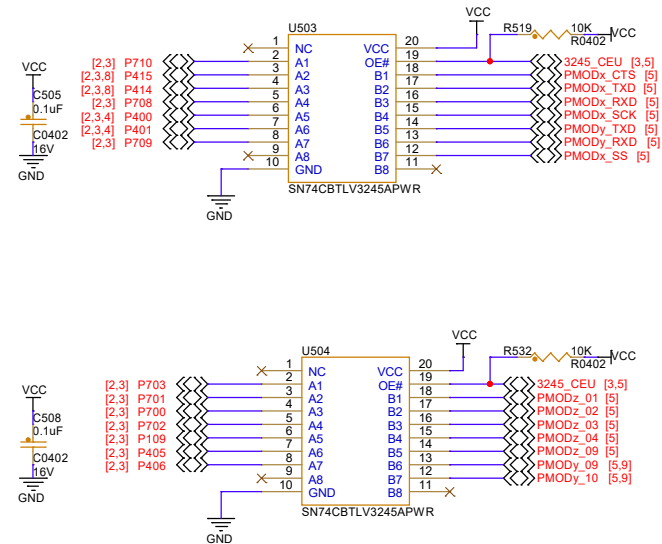
PMOD使用立贴母座，
横向间隔中心距9x2.54mm
纵向间隔中心距4x2.54mm

电平转换

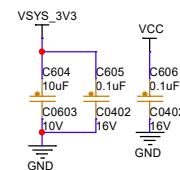
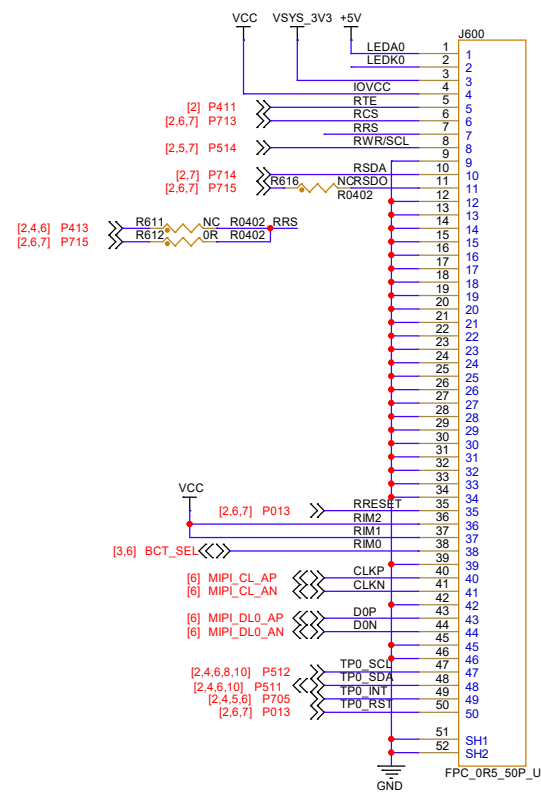
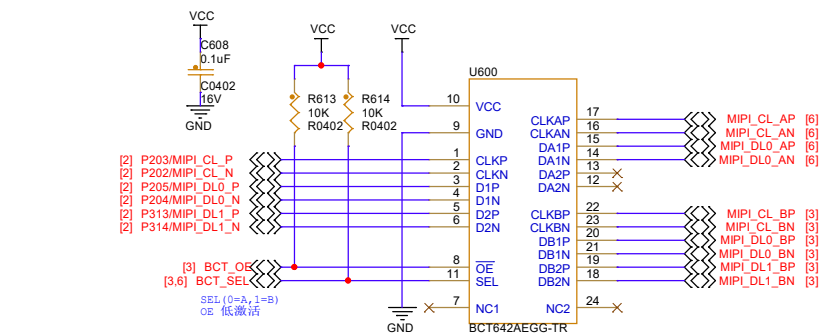
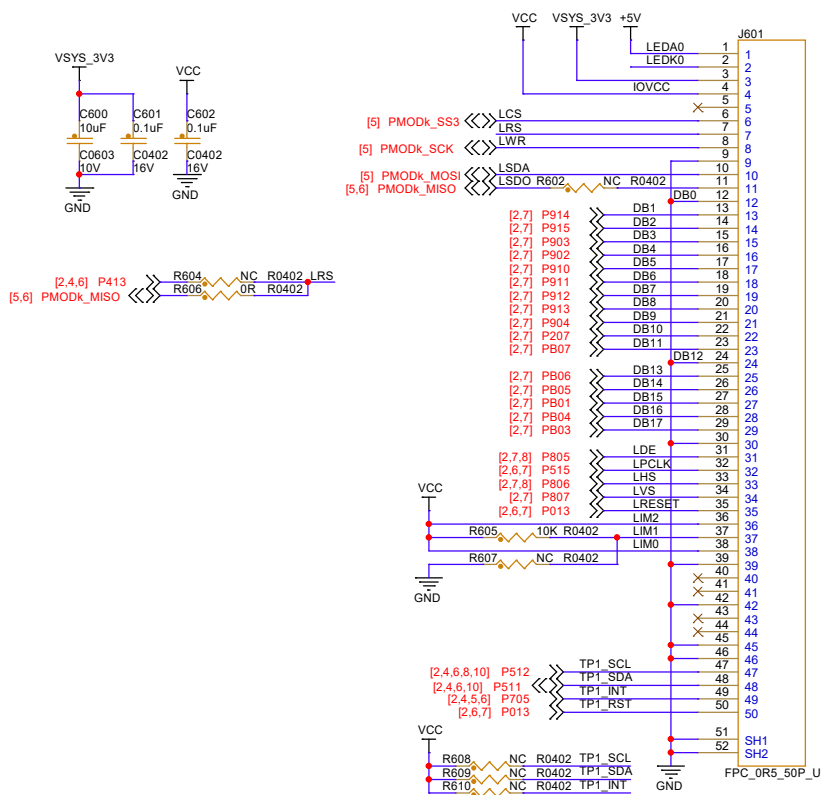


Page1: USB_FS
Page2: BTB-
Page3: OV/MIPI
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Page7: RGB转HDMI/HDMI接口
Page8: 千兆网口
Page9: 音频PDM
Page10: Arduino

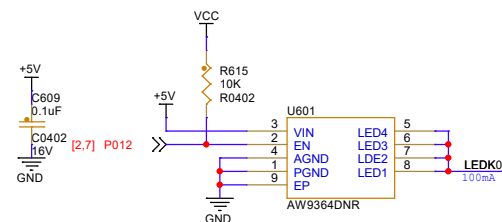
电路开关



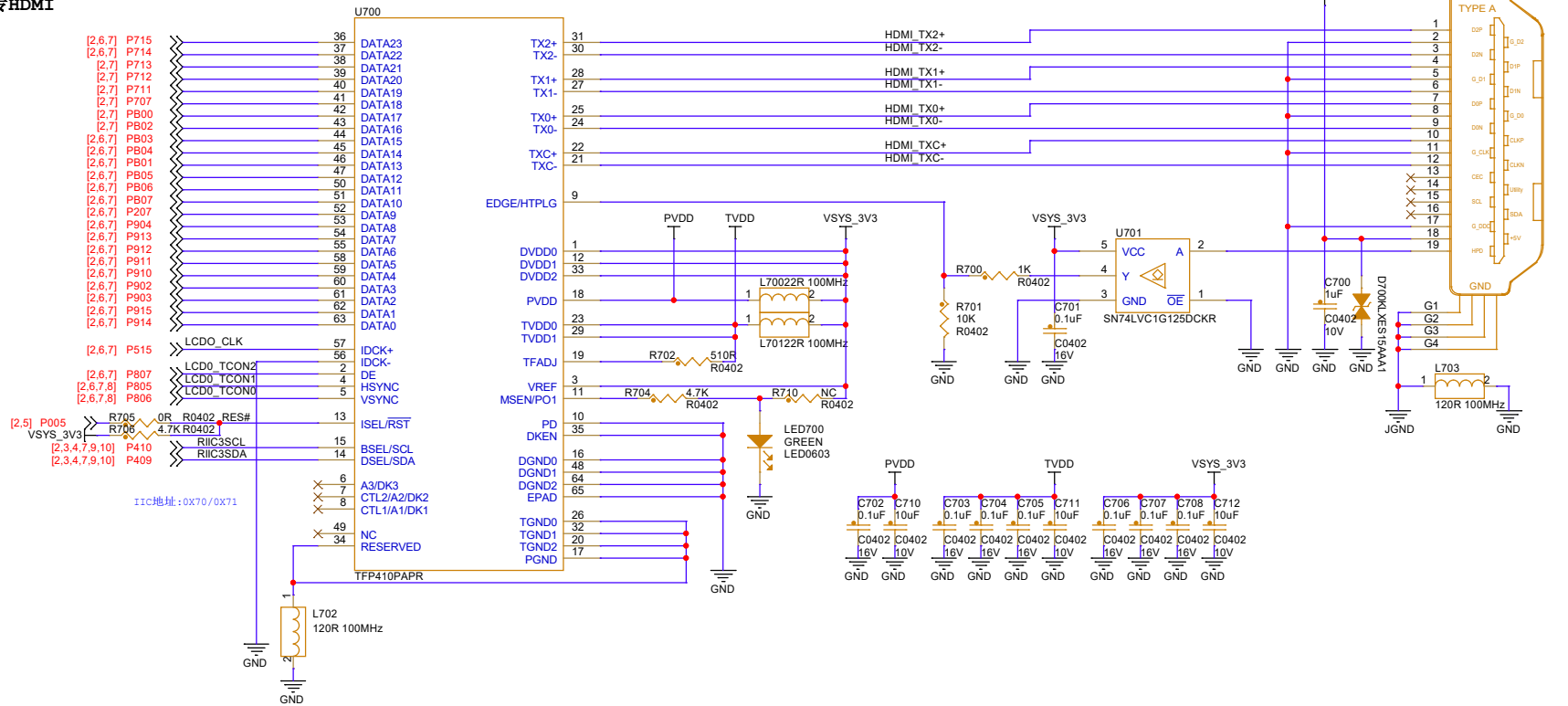
[2.5,10] PC14_P603
[2.5,10] PC13_P602
[2.5,8] PC11_P604



IM2	IM1	IM0	interface mode	data pin
0	0	0	18 bit parallel	DB17:DB0
0	0	1	9 bit parallel	DB08:DB0
0	1	1	16 bit parallel	DB15:DB0
0	1	1	8 bit parallel	DB7:DB0
1	0	1	3SPI	SDA :in/out
1	1	0	MPPI	MIPPI_DATA
1	1	1	4SPI	SDA :in/out



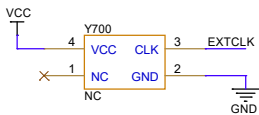
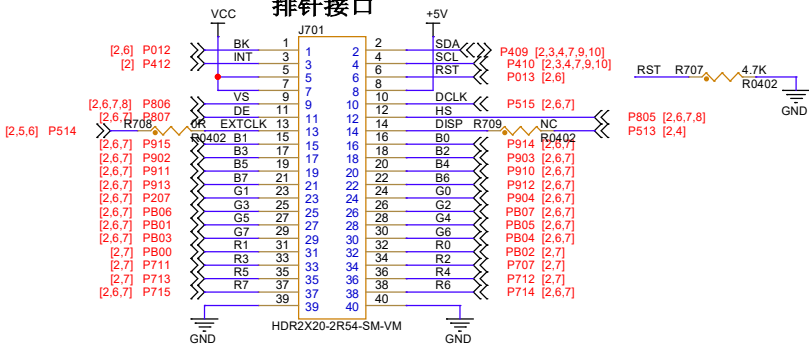
RGB转HDMI



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 Page3: OV/MIPI
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5.0 TFT 800*480

排针接口



以太网

TXC / RXC : delays 2ns
PHY Address : 001

[2] P906
[2] P907
[2] P908
[2] P909
[2] P905
[2] P206

[2] P307
[2] P306
[2] P305
[2] P304
[2] P310
[2] P309

[8] RGMII_MDC
[8] RGMII_MDIO
[2,4,6,10] P512
[2,5] P706
[2] P308

[2] PA07

U800
DVDD_RG
RSET
AVDD33
AVDD33_1
DVDD33
DVDD10
AVDD10
AVDD10_1
AVDD10_2
REG_OUT
GND
MDIP0
MDIN0
MDIP1
MDIN1
MDIP2
MDIN2
MDIP3
MDIN3
PHY_LED0
PHY_LED1
LED2
RTL8211F

MDC
MDIO
INTB
CLKOUT
XTAL_OUT
XTAL_IN
PHYRSTB

Y800
25MHz

LED0
LED1
LED2

RTL8211F

3V3_DVDD_PHY

LED_Y+

LED_G+

LED0

PHY_LED0

PHY_LED1

PHY_LED2

PHY_LED3

PHY_LED4

PHY_LED5

PHY_LED6

PHY_LED7

PHY_LED8

PHY_LED9

PHY_LED10

PHY_LED11

PHY_LED12

PHY_LED13

PHY_LED14

PHY_LED15

PHY_LED16

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PHY_LED283

PHY_LED284

PHY_LED285

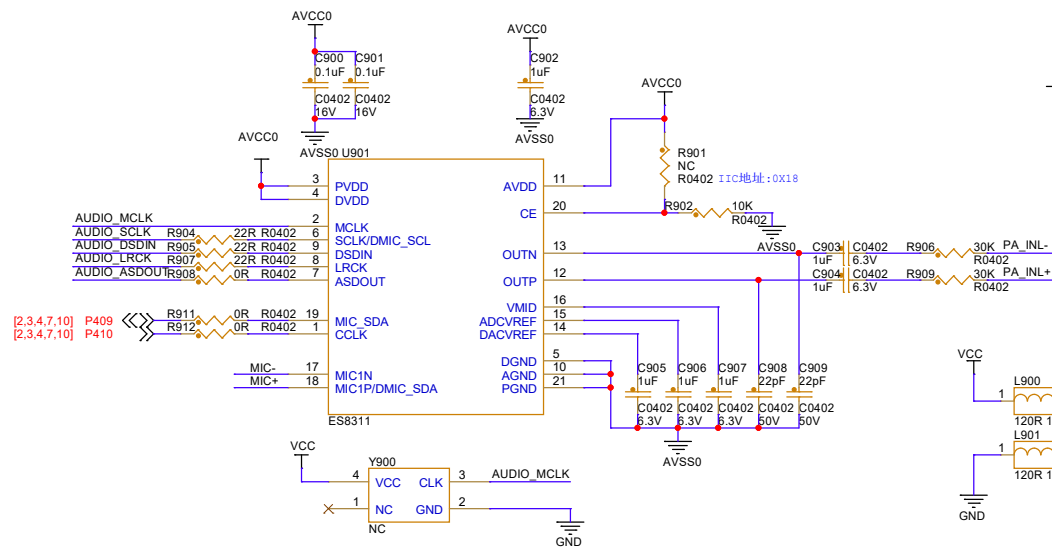
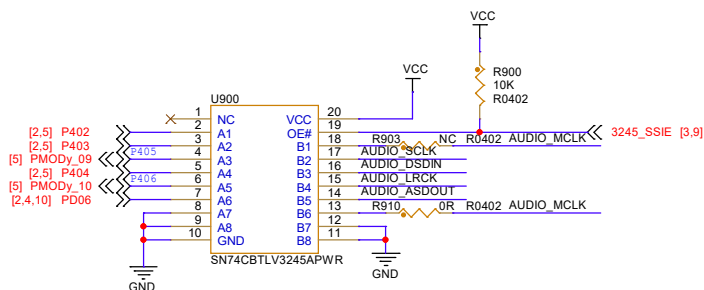
PHY_LED286

PHY_LED287

PHY_LED288

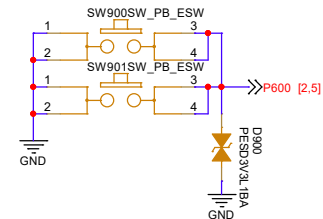
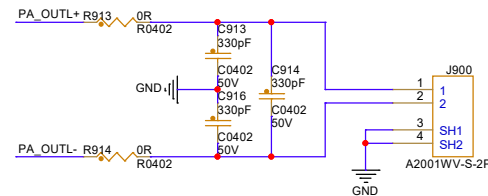
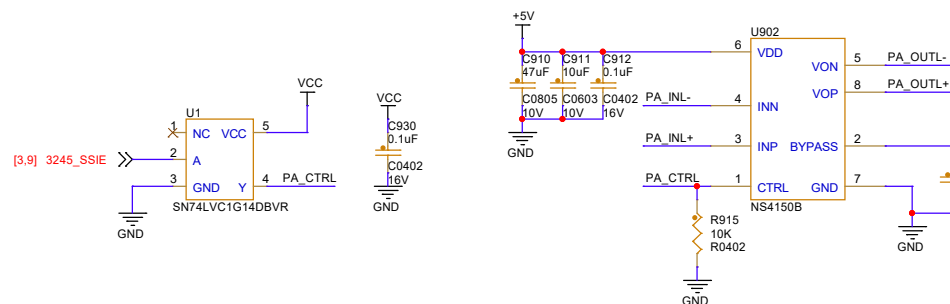
PHY_LED289

PHY_LED290

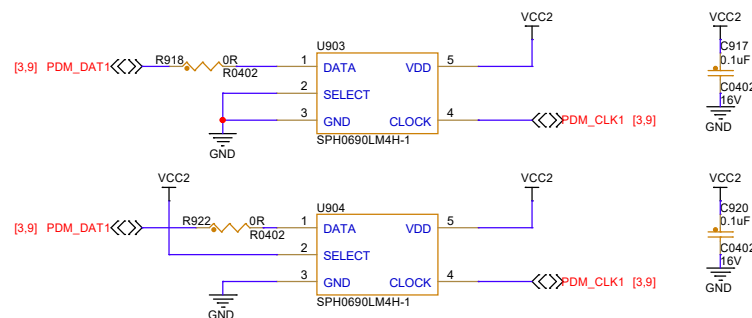


Page1: USB_FS
Page2: BT
Page3: OV/MIP1
Page4: CAN/陀螺仪
Page5: PMOD
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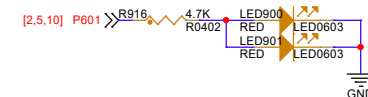
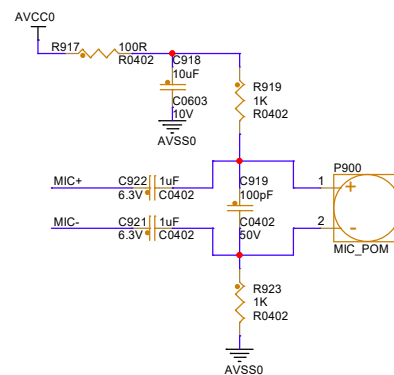
SPERK



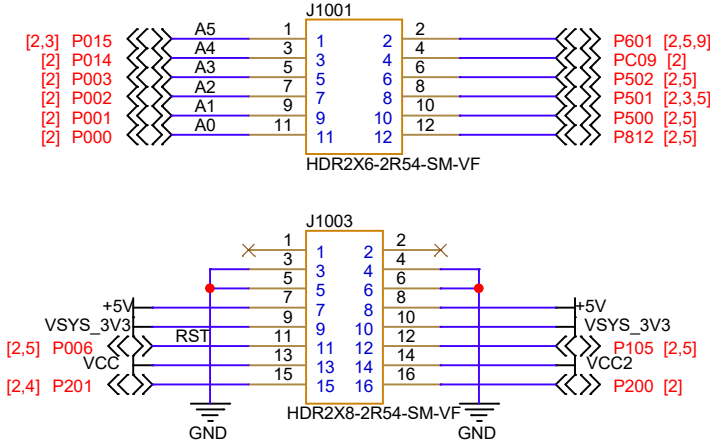
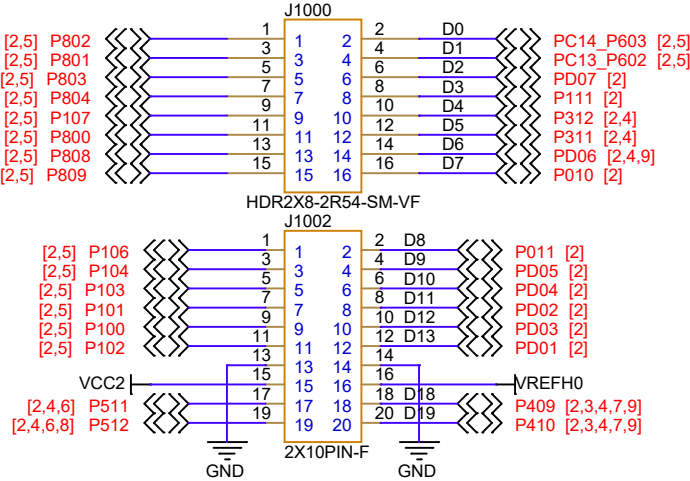
PDM



MIC



Arduino



54321

D

CPKEXP-EK8x2扩展板_原理图_v1.0

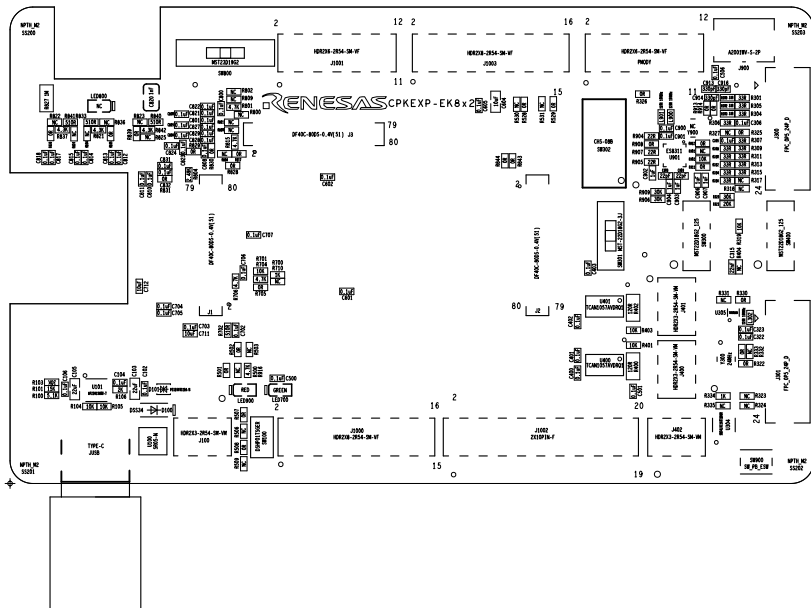
C

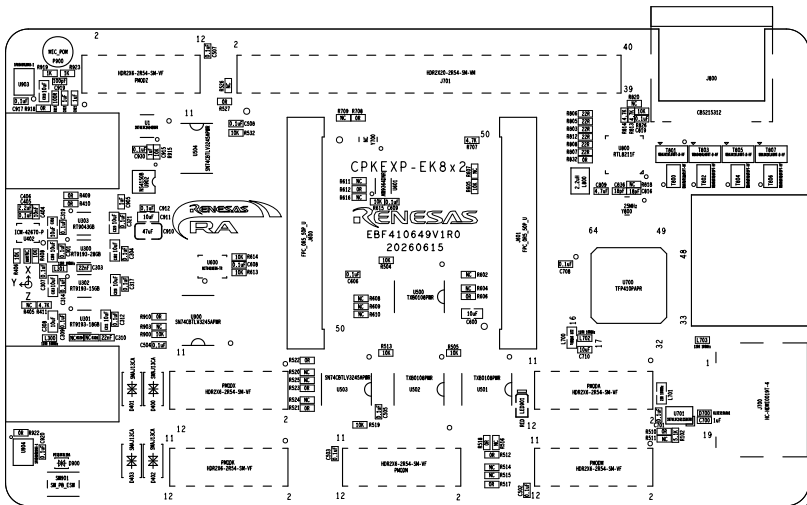
目录

Page 1	USB_FS
Page 2	BTB
Page 3	OV/MIPI
Page 4	PDM/CAN
Page 5	PMOD
Page 6	RGB_LCD
Page 7	RGB转HDMI/HDMI接口
Page 8	千兆网口
Page 9	音频/陀螺仪
Page 10	Arduino
Page 11	目录
Page 12	历史版本

历史版本

版本号	日期	设计	描述
V1.0	2026-06-09	llm	初版发行





CPKEXP-EK8x2 Carrier Board Power Supply

